

The Heritage School
Pre-board Examination 2016 - 17
Class – XII
CHEMISTRY

Full marks – 70

Time – 3 hrs

(Candidates are allowed additional 15 minutes for only reading the paper. They must NOT start writing during this time)

Answer all questions in **Part I** and six from **Part II**.

All working, including rough work, should be done on the same sheet as and adjacent to, the rest of the answer.

The intended marks for questions or parts of questions are given in brackets [].
Balanced equations must be given wherever possible and diagrams where they are helpful.

When solving numerical problems, all essential working must be shown. In working out problems, use the following data.

Gas Constant $R = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 0.0821 \text{ dm}^3 \text{ atm K}^{-1} \text{ mol}^{-1}$. $1 \text{ L atm} = 1 \text{ dm}^3$

PART – I

(Answer all questions)

Question – 1 :

a) **Fill in the blanks with appropriate words given below :**

[5]

[Greater, lesser, nitric acid, nitrous acid, common ion effect, salt hydrolysis, decreased, increased, does not change, acidic, basic, neutral, PCl_5 , SOCl_2 , methyl cyanide, primary amine, vapour, vapourpressure, positive, negative]

- i) Solubility of Calcium oxalate is _____ in the presence of ammonium oxalate because of _____.
- ii) Ethanol is _____ to litmus while phenol is _____ to litmus.
- iii) The _____ pressure of an aqueous solution of 0.1(M) cane sugar is _____ than that of pure water measured at the same temperature.
- iv) On treating a _____ with _____ nitrogen gas is evolved.
- v) The more _____ the reduction potential of a metal, the _____ is its ability to displace hydrogen from acids.

b) Match the following :

[5]

A		B	
i)	Nernst equation	a)	Glycine
ii)	Ostwald's dilution law	b)	Ammonia
iii)	Lewis base	c)	Amine
iv)	Zwitter ion	d)	Electrode potential
v)	Hinsberg's reagent	e)	Weak electrolyte

c) Explain the following facts in brief :

[5]

- Unlike most phenolic compounds, 2,4,6-trinitrophenol is soluble in sodium bicarbonate solution.
- Specific conductance of a solution decreases with dilution.
- Transition metals form complex compounds.
- Interhalogen compounds are more reactive than their constituent elements.
- Zinc sulphide is precipitated from an ammoniacal solution in presence of hydrogen sulphide.

d) Choose the correct answer from the alternatives given :

[5]

- The number of unpaired electrons in the complex ion $[\text{CoF}_6]^{3-}$ will be :
 (a) 2 (b) 3 (c) 4 (d) 0
- The fibre obtained by condensation of hexamethylenediamine and adipic acid is :
 (a) dacron (b) nylon 66 (c) rayon (d) teflon
- In the metallurgy of copper, matte is considered to be a mixture of :
 (a) $\text{FeS} + \text{ZnS}$ (b) $\text{CuS} + \text{FeS}$
 (c) $\text{CuFeS}_2 + \text{FeS}$ (d) $\text{Cu}_2\text{S} + \text{FeS}$
- Wohler's synthesis is used for the preparation of:
 (a) glycine (b) amino acids (c) urea (d) protein
- Dehydration of amide with P_2O_5 yields :
 (a) NH_3 (b) RCN (c) RNC (d) RNH_2

Part – II
Section A

Question – 2 :

- a) (i) A solution containing 8 gm of compound per 100 ml solution was found to have an osmotic pressure of 500 cm of Hg at 27°C. calculate the molecular mass of the compound. [2]
- (ii) What happens to the boiling point of water when small amount of sodium chloride is dissolved in it? [1]
- b) (i) The density of A^+B^- which has (CsCl) bcc structure is 4.4 gm/cm^3 . Calculate the edge length of the crystal. [Formula mass = 213] [2]
- (ii) What is the coordination number of copper in its crystal? [1]
- (iii) An aqueous solution of acetic acid and sodium acetate having PH = 4.2 is diluted to half the initial concentration. What will be the PH of the new solution. [1]
- c) (i) If one Faraday were to be 48250C instead of 96500C, what will be the charge of an electron? [2]
- (ii) Explain the term Ferrimagnetic. [1]

Question – 3 :

- a) (i) The values of λ_m^∞ for HCl, NaCl and CH_3COONa are 426.1, 126.5 and $91 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. Calculate the value of λ_m^∞ for acetic acid. [2]
- (ii) The temperature coefficient for the hydrolysis of ethyl acetate by sodium hydroxide solution is 1.75. Calculate the energy of activation for the reaction. [3]
- b) Deduce the equation to calculate the PH of an acid buffer solution. [2]
- c) AgNO_3 solution cannot be stored in a copper vessel – Explain. [1]
- d) Calculate the molality of solution prepared by dissolving 3.6 gm glucose in 500 gm water. [2]

Question – 4 :

- a) The standard electrode potential of Pb/Pb^{2+} and $\text{Pt/I}^-/\text{I}_2$ are -0.126 v and 0.536 v respectively. When a galvanic cell is constructed using 0.1 M concentration of the ions. Pt is found to be the cathode. What is the emf of the cell? [3]
- b) (i) What is Arrhenius equation? How can it be used graphically to calculate the energy of activation for a reaction? [2]
- (ii) The solubility product of BaSO_4 is 1.5×10^{-9} . Find its solubility in 0.1 M BaCl_2 solution. [2]

- (c) (i) How many formula units of sodium chloride are present in an unit cell of sodium chloride?
(ii) The what type of cubic system does this cell belong?
(iii) What is the geometry of sodium ions around chloride ions? [3]

SECTION B

Answer any two questions

Questions – 5 :

- a) Write the formulae of the following coordination compounds:
(i) Potassium dicyanoargentate(I) (ii) hexaaquairon (III) chloride [2]
b) Using C.F.T determine the number of unpaired electrons for FeCl_4^- complex [1]
c) Explain why tetrahedral nickel complexes are paramagnetic but square planar Ni(II) complexes are diamagnetic. [2]

Question 6 : -

- a) Explain the following facts:
i) Silver bromide is used in photography.
ii) In a given transition series, there is no significant change in the atomic radii of elements with increase in atomic number. [2+2=4]
b) Draw the structure of xenon oxyfluoride molecules [1]

Question 7 : -

- a) Give balanced equations for the following reactions.
(i) Ozone and mercury.
(ii) Chlorine is passed through hot concentrated NaOH solution.
(iii) Silver nitrate is added to dilute solution of sodium thiosulphate [3]
b) How is silver extracted from its sulphide ore by cyanide process? Give only the reactions. [2]

Section – C

(Answer any two questions)

Question – 8 :

- a) How can the following conversions be brought about:
- (i) Benzoic acid to phenol [2]
 - (ii) Acetaldehyde to acetamide. [1]
 - (iii) Aniline to fluorobenzene [1]
 - (iv) Propionic acid to acetic acid [2]
- b) Identify the products A,B,C and D [2]
- $\text{CH}_3\text{CH}_2\text{Br} \xrightarrow{\text{KCN}} \text{A} \xrightarrow{\text{H}_2\text{O}/\text{H}^+} \text{B} \xrightarrow{\text{PCl}_5} \text{C} \xrightarrow{\text{NH}_3} \text{D}$
- c) Complete the given reactions [2]
- (i) $\text{CH}_3\text{COCH}_3 + \text{I}_2 + \text{KOH} \longrightarrow$
 - (ii) $\text{C}_6\text{H}_5\text{NH}_2 + \text{CH}_3\text{COCl} \xrightarrow{\text{Pyridine}}$

Question – 9 :

- a) Give balanced equations for the following name reactions: [3]
- (i) Kolbe-Schmidt reaction.
 - (ii) Hellvolard's Zelinsky reaction
 - (iii) Clemmensen's reduction reaction.[
- b) An organic compound[A] with molecular formula $\text{C}_3\text{H}_6\text{O}$ which reduces Tollen's reagent gives compound [B] on treatment with hydroxylamine. [B] on reduction with LiAlH_4 gives compound [C]. Compound [C] on warming with chloroform and potassium hydroxide gives compound [D] with unpleasant smell. [D] on reduction with $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ gives [E] which gives insoluble compound with benzene sulphonyl chloride and NaOH . Give the sequence of reactions and suggest structures of[A] to [E] [4]
- c) Give one chemical test to distinguish between the following pairs. [3]
- i) Formic acid and oxalic acid .
 - ii) Glucose and fructose

Question – 10 :

- a) Write balanced reactions for the following: [4]
- (i) Acetamide is heated with phosphorous pentoxide followed by hydrolysis.
 - (ii) Benzoic acid is treated with a mixture of conc. H_2SO_4 and conc. HNO_3 .

- (iii) Ethyl magnesium chloride is treated with formaldehyde and the product is hydrolysed in presence of an acid.
- (iv) Diethyl ether reacts with excess of HI.
- b) On electrolysis in acidic solution amino acids migrate towards cathode, while in alkaline solution these migrate towards anode. Explain [2]
- c) Deficiency of which vitamin will cause the following diseases: [2]
(i) Night blindness (ii) Ricket
- d) Name the type of polymerization and the monomers in each of the following polymers:
(i) Dacron (ii) Nylon- 66 [2]